Total No. of Questions : 10] [Total No. of Printed Pages : 4

Paper Code : 21316

F-416

B.C.A. (Fourth Semester)

Examination, 2025

(New Course)

Paper -BCA-(N)-401

Operating System

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks. Symbol used have their usual meanings.

1. (a) Discriminate between batch operating system and multiprocessor operating system with neat and clean diagram.

   (b) What do you mean by non-preemptive algorithm? Explain it through an example.

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2. (a) Define deadlock. Also Describe deadlock detection methods.

   (b) Write short notes on the following terms:

   (i) Virtual memory

   (ii) FCFS

3. (a) Compare long term, short term and midterm scheduler.

   (b) List the various components of Operating System.

4. (a) Consider the following reference string:

   1, 2, 3, 4, 2, 1, 6, 5.1.2.3,7.6, 3, 2.1.2, 3.6

   How many page faults will occur for:

   (i) LRU replacement

   (ii) Optimal page replacement

   (b) Write the name of disk schedule algorithm. Write the method and explain the working of any two algorithm.

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5. (a) What is Fragmentation Problem? Describe the external and internal fragmentation.

(b) What the basic functions does an operating system perform as a resources manager?

6. (a) What is Process Control Block? Design basic framework of process control block.

(b) Explain in details about the disk storage and disk scheduling.

7. (a) Explain in detail about the mutual exclusion and critical section problem.

(b) For the following processes, draw Gantt Chart to illustrate the execution using,

(i) SJF

(ii) Round Robin

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Also, calculate average waiting time and average turnaround time. (Assumption: A larger priority number has higher priority.

Process	Arrival time	Burst time	Priority
A	0	5	4
B	2	4	2
C	2	2	6
D	4	4	3

8 (a) Explain in detail about the inter process communication and models.

(b) Differentiate between Best fit and Worse fit.

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